

ACHINTYA RAI

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EDUCATION

University of California, San Diego
Bachelor of Science in Computer Engineering

Sept. 2024 – June 2028
GPA: 3.9/4.0

WORK EXPERIENCE

Adobe

Software Engineering Intern

San Jose, CA

June 2026 – Sept. 2026

- Designed a machine learning anomaly detector using statistical algorithms to flag payment anomalies across **30+** countries and **10+** large-scale workflows.
- Applied software design principles to build an MLOps pipeline for weekly retraining and hourly inference; surfaced a **26-hour** outage across **1,272** monitoring segments.
- Profiled **60.6M+** payment events in Databricks, identifying **86,387** duplicate records and a **47%** null rate to establish the primary validation benchmark.

VALT Health

Founding Software Engineer

Remote

Aug. 2025 – Feb. 2026

- Leveraged concurrency to build large-scale systems in GCP synchronizing **100k+** fitness datapoints, achieving a **35%** speed increase and **60%** reduction in database writes.
- Engineered a Flask backend unifying formats across **20+** sources (Fitbit, Apple Watch), cutting test setup by **80%** via automated Bash scripts.

RESEARCH

Graph Theory & Neural Architecture

Undergraduate Research Fellow (NSF REU)

San Diego, CA

March 2026 – Present

- Engineered a Python pipeline to analyze LLM embedding manifolds (Llama-2-70B), applying custom data structures and algorithms to compress **500GB** checkpoints to **13GB** localized manifolds.
- Optimized preprocessing scripts to achieve a **30x** runtime reduction, enabling large-scale manifold analysis on resource-constrained **8GB RAM** systems.

Data Science Teaching & Learning Lab

Undergraduate Researcher

San Diego, CA

Sept. 2024 – Present

- Developed a scalable LLM-augmented tool using the Gemini API to automate labeling of **16k+** AI usage data points, reducing labeling time by **90%**.
- Architected a multi-agent artificial intelligence tutoring framework with hallucination-reduction strategies, achieving **95%+** accuracy across **100k+** data points.

PROJECTS

NanoGPT | *Python, PyTorch, C++, Transformers, LLM Optimization* | [🔗](#)

Mar. 2026

- Implemented a GPT-style transformer with machine learning algorithms for multi-headed attention and backpropagation, sustaining **100%** hardware utilization during scalable training.
- Compiled high-performance C++ sub-modules and optimized memory layout for distributed systems inference, reducing per-token latency by **25%** on resource-constrained hardware.

Watchdog | *Python, Gemini API, Browser-Use, Asyncio, SQLite* | [🔗](#)

April 2026

- Engineered AI-integrated software using Gemini Vision to autonomously monitor and intervene on browser tabs, delivering a functional agent system at DiamondHack 2026.
- Built a concurrent CLI with Asyncio for real-time URL blacklisting and Notion API synchronization, eliminating context-switching overhead across active browser sessions.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, TypeScript, SQL (Postgres/SQLite), Bash, HTML/CSS

Development: React, Django, FastAPI, Asyncio, WebSockets, Flask, Node.js, JUnit, PyTest

AI & Cloud: PyTorch, AWS, GCP, Gemini/OpenAI API, Docker, GitHub Actions, LLMs, Linux